Python Programming Foundations

Day 1 - Coding/Course Setup

"Creating the working environment"

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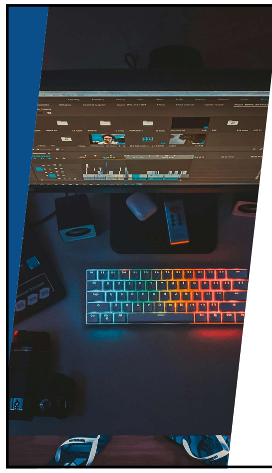
Setup Is Part Of The Course



Before writing Python comfortably, your computer needs a stable setup.

The goal is not to memorize setup steps. The goal is to create a reliable place to run your code.

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Setup Path

Course setup path:

- Python ->
- project folder ->
- `.venv` ->
- VS Code ->
- GitHub

Each piece supports a different part of the workflow.

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What Success Look Like

- check your Python version
- open a project in VS Code
- select a `.venv` interpreter
- run a small `.py` file
- see your code in GitHub or prepare it for submission

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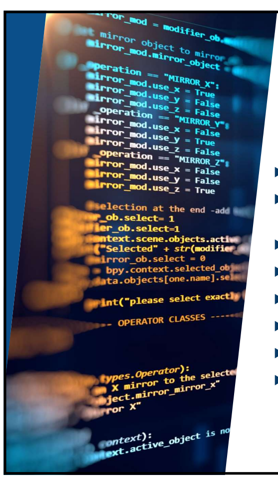

What We Will Use Today

- Python from `python.org`
- PowerShell or terminal verification commands
- project folders
- `.venv` virtual environments
- Visual Studio Code
- Microsoft Python extension
- VS Code interpreter selection
- VS Code source control / GitHub integration

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What We Will "Skip" For Now

- advanced dependency management
- publishing packages
- complex Git branching
- pull request reviews
- deployment
- Python pre-release versions

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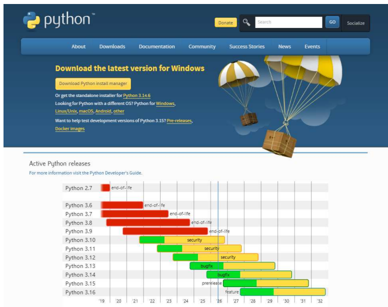
Download Python



url: python.org/downloads

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Windows – Two options

Either option will work
(The top one is the easiest)

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New: Python Install Manager

(Your textbook does not mention this)

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Install Python

- The Install manager will make sure that a usable `py` or `python` command is created.
- If the installer offers PATH-related options, you should enable the option that lets Python run from the terminal.
- Once the process is finished:
  - Open a terminal (powershell)
  - On the command line type `py`
  - Optionally, next type the command `help'

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Verify Python

- Once the process is finished:
  - Open a Terminal or Powershell and run one of the following commands:
    - `python --version`
    - `py --version`
    - `python -m pip --version`
  - Optionally, launch the interactive python shell and try the help command
    - `py`
    - Then type: `help`
    - Within help: `keywords`, `modules`, `symbols` or `topics`

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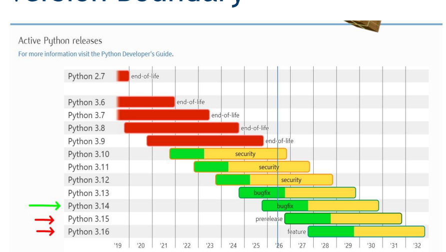
Version Boundary



- Use a ***stable*** Python release.
- Do not use pre-release versions for this course unless the instructor tells you to.
- *"You don't want to get 'bit'!"* 😝

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Create A Course Folder

Create a folder, or folder structure for your course work.

Keep your projects and assignments organized from the start.

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What A Virtual Environment Does

- A virtual environment is a local Python workspace for one project.
- It helps keep project packages separate from the rest of your computer.
- It also allows for different “versions” of python to “co-exist” on one machine. This will be important when working on legacy software applications.

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Create `.venv`

From the project folder, using the terminal or powershell, create a virtual environment:

- `py -m venv .venv`

If `py` is not available, use:

- `python -m venv .venv`

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Activate `.venv` In PowerShell

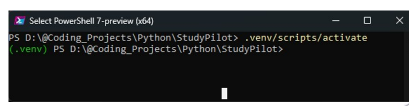
Windows PowerShell activation:

Command:

- `.\.venv\Scripts\Activate.ps1`

When active, the terminal prompt usually shows:

- (.venv)



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Why `.venv` Matters Later

Later: when projects use packages, `.venv` helps keep installs project-specific.

For Week 1: the goal is simply to know what `.venv` is and how to create/select it.

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Install Visual Studio Code

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Difference between VS Code and Visual Studio

Visual Studio is a true IDE - Integrated Developer Environment.

- Heavy footprint (takes a lot of memory)
- Great for C# and .NET framework development
- Doesn't play well on an iOS machines

VS Code is a code editor, not a true IDE ... but can function as a full-IDE by installing the extensions

- Widely used!!
- Smaller footprint (faster load and execution)
- Updated very frequently
- Supports multiple development languages and frameworks

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Download VS Code

url: code.visualstudio.com/download

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VS Code Extensions

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Install Python Extensions

Recommended VS Code extensions:

- Python
- Pylance
- Python Environments
- Python Extension Pack, optional (Don Jayamanne)
- Python Development Extensions Pack, optional
- Jupyter, optional

Use trusted publishers such as Microsoft.

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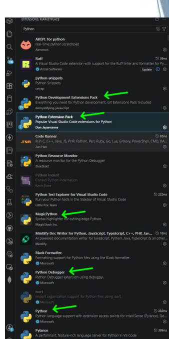
Install Python Extensions

Select the Extensions button on the left side bar.

Search for the desired extensions

Click on the "Install"



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Install Optional Additional Extensions

Some cool extensions

- Better Comments
- Better Todo Tree
- Bracket Pair Color DLW
- Colorful Dark+
- Colorful Comments
- #region folding for VS Code
- Powershell Extension Pack
- Print

- Git Extensions Pack
- Git Graph

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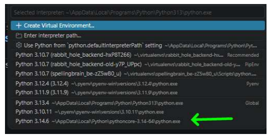
Select The Python Interpreter

In VS Code:

1. Open the project folder.
2. Open the Command Palette (Ctrl+Shift+P).
3. Choose `Python: Select Interpreter`.
4. Select the interpreter inside `.venv`.



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VS Code And `.venv`

VS Code can detect and use virtual environments.

When the `.venv` interpreter is selected, new VS Code terminals can activate that environment automatically.

If it is not selected, you will need to manually activate the virtual environment using the activate command

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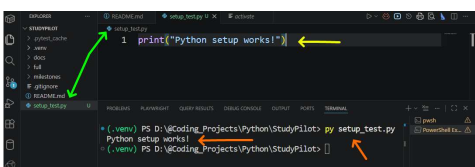
Run A Test File

Create a file: `setup_test.py`

Add the line: `print("Python setup works!")

Execute the file from integrated terminal: `py setup_test.py`



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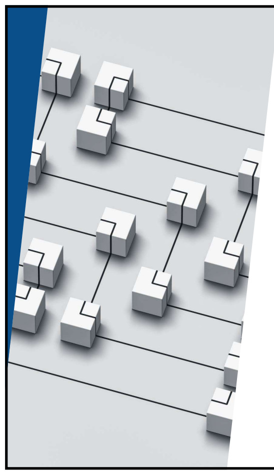

GitHub Submission Paths

There are two common GitHub paths:

- Git CLI commands *(you should learn these regardless)*
- VS Code Source Control and GitHub extensions

Both paths should preserve code changes clearly.

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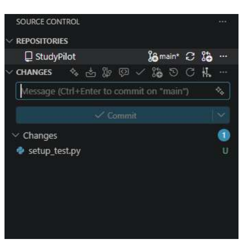
Source Control In VS Code

VS Code has a Source Control view for Git work.

You can see changed files, write a commit message, commit, and sync without typing every Git command manually.



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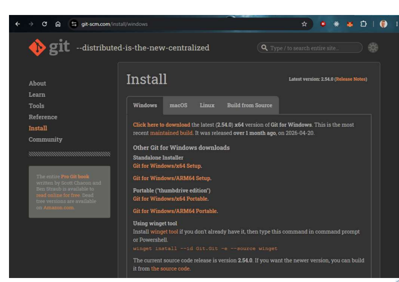
Git Install for Windows



url: git-scm.com/install/windows

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Initialize git for your project(s)

Using either VS Code or the CLI – initialize git (locally) for your project.

- `git init`

Note:

- This does not setup a GitHub repository in the cloud. It only sets it up local.
- Later we will connect the local with the GitHub repo

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GitHub Extensions

Useful GitHub-related VS Code extensions:

- GitHub Pull Requests
- GitHub Repositories

These can help with GitHub sign-in, repositories, commits, and pull-request workflows.

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GitHub Vocabulary

Core GitHub Words

Keyword	Definition/Use
Repository	Project storage
Clone	Copy repository to your computer
Commit	Save a meaningful change
Push	Send commits to GitHub
Sync	Pull and push changes through VS Code

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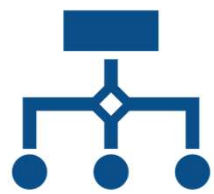
CLI Or VS Code

Git CLI and VS Code can both support the same workflow:

change files ->
review changes ->
commit ->
push or sync

Use either path (or the one your instructor approves) for the assignment.



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Setup Verification Checklist

Setup checklist:

- Python version command works
- project folder exists
- `.venv` exists
- VS Code opens the folder
- VS Code selects the `.venv` interpreter
- `setup_test.py` runs
- GitHub workflow is understood or ready to practice

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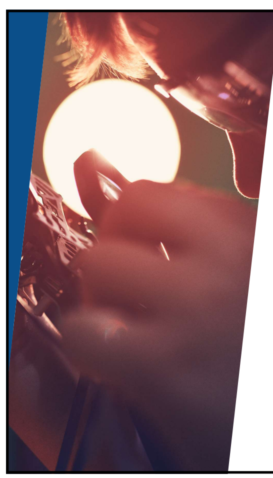

Troubleshooting Layers

When setup fails, locate the layer:

- Python install
- terminal command
- project folder
- `.venv`
- VS Code interpreter
- Git or GitHub sign-in

Fix one layer at a time.

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Ready For First Programs

You are ready for Week 1 coding when:

- Python runs
- VS Code opens your project
- `.venv` is available
- a tiny `.py` file prints output
- your submission path is clear

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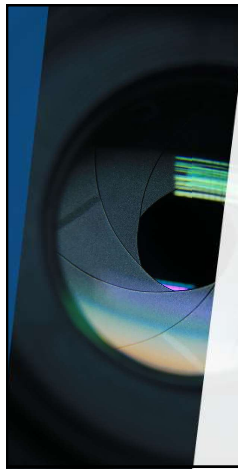

Demo/ Walkthrough

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Demo / Walkthrough Notes (Review)

1. Open `python.org/downloads/`.
2. Show the latest stable Python release.
3. Install or verify Python.
4. Run version checks in PowerShell.
5. Create a project folder.
6. Create `.venv`.
7. Open VS Code.
8. Install Python-related extensions.
9. Select the `.venv` interpreter.
10. Run `setup_test.py`.
11. Source Control and GitHub extension locations in VS Code.

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Your Turn

Install Python

Run version checks

Install VS Code if needed

Setup Project Folders

Create .venv

Open VS Code

Install Extensions

Select the interpreter

Run `setup_test.py`

Initialize git locally

Commit your change(s)

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Setup Evidence

Recommended evidence:

- screenshot or copied terminal text showing Python version
- screenshot or note showing `.venv` exists
- screenshot or copied output from `setup_test.py`
- confirmation that VS Code selected the `.venv` interpreter
- confirmation of local git, GitHub sign-in, or instructor-approved submission path

Keep this as setup evidence only or for troubleshooting.

This is NOT graded.

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Great!

Now let's start the class!

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